PayGuard - Part C: Production Design Memo

Recommendation

Pilot the Logistic Regression fraud detection model as a human-reviewed alert system. Do not automatically block transactions at this stage. The complaint triage prototype needs more evaluation because only 8 of 10 attempted complaints were successfully processed in the small pilot.

Evidence and trade-off

On the held-out test set, Logistic Regression at threshold 0.95 detected 40 of 52 fraud cases (76.92% recall). It produced 55 false alerts, missed 12 fraud cases, and achieved 42.11% precision and 70.69% average precision. This supports a controlled pilot, not full automation. The threshold was chosen on validation data to reduce review workload, and should be revisited using actual fraud losses and review costs.

Proposed workflow

When a card transaction arrives, the payments system prepares the same input features used in training and passes them to the saved model. The model returns a fraud score. Scores at or above 0.95 create an alert for the fraud review team. A reviewer checks the transaction and decides what action to take. Record the score, alert, review outcome, and confirmed fraud label when available. Normal payment processing should not depend on a reviewer responding instantly.

Requirements before integration

The Kaggle dataset uses anonymized PCA features V1-V28. PayGuard must confirm that equivalent features can be generated consistently for live transactions. The dataset Time field is elapsed time from the original data collection and cannot be copied directly into a live system without redesign. Feature order and scaling must match training; no future information should enter the prediction.

Monitoring and silent failures

Track alert volume, precision, recall once confirmed labels arrive, missed fraud, false positives, review backlog, prediction latency, and input-data changes. Also watch for missing or incorrectly ordered features, model/API errors, and unexpected drops in alert volume. These may indicate a silent failure even if the system is still running. Keep a manual-review fallback when scoring fails.

Privacy and data handling

Limit access to transaction details and fraud scores to authorized staff. Encrypt data in transit and at rest, avoid unnecessary personal data in logs, and set retention periods with the compliance team. Confirm applicable data-residency requirements before using external infrastructure. Do not send raw customer financial data to third-party services without approval.

Phased rollout and human oversight

Phase 1: Validate live feature availability and test the model offline. Phase 2: Run in shadow mode without affecting payments, comparing alerts with confirmed outcomes. Phase 3: Send a limited number of alerts to human reviewers. Phase 4: Expand only if fraud detection, false-alert workload, and reliability are acceptable. Keep human review for flagged cases throughout the pilot.

Dependencies and decision requested

Engineering: live features, scoring integration, logging and fallback. Fraud/Risk team: review workflow, fraud labels and acceptable alert volume. Security/Compliance: access, retention and residency approval. Request: Approve a small, monitored human-review pilot after these dependencies are checked. Do not approve automatic blocking yet.